Journal of Materials Research and Technology 32 (2024) 439–451

Contents lists available at ScienceDirect

Journal of Materials Research and Technology

journal homepage: www.elsevier.com/locate/jmrt

Enhancing strength and ductility in Fe–20Mn–9Al-1.5C austenitic low-density steels through Ni and Cr alloying synergy adding

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ARTICLE INFO

Handling editor: P Rios

Keywords:
Austenitic lightweight steels
Ni & Cr alloyed
Mechanical properties
Deformation mechanism

ABSTRACT

In this investigation, a Fe–20Mn–9Al-1.5C lightweight austenitic steel has been developed through strategic alloying with Ni and Cr elements coupled with the employment of a thermo-mechanical processing technique featuring ultra-fast cooling. This composition and treatment synergistically transcend the conventional trade-off between strength and ductility, positioning the mechanical performance of this steel notably above that of comparable low-density austenitic steels previously reported. The coordinated incorporation of Ni and Cr is key to modulating the precipitation threshold for nanoscale κ-carbides and LRO domains, achieving an optimal size for precipitate-induced strengthening. The incorporation of Cr specifically contributes to a dual enhancement mechanism: it fosters the generation of high-density dislocation arrays (HDDA) and refines the dynamic slip band (DSBR) during deformation, which in turn, elevates the strain hardenability and ductility. The remarkable synergy in strength and ductility is attributed to a combination of mechanisms including grain boundary reinforcement, solid solution hardening, the precipitation strengthening effect of nanoscale precipitates, and the activation of secondary slip system dislocations. This synergy is quantitatively reflected in the material's yield strength (YS) exceeding 1100 MPa, ultimate tensile strength (UTS) surpassing 1200 MPa, and a total elongation (TEL) exceeding 45%. The insights gained from this research provide critical guidance for the design and creation of austenitic low-density steels that do not compromise between strength and ductility, thereby offering superior overall performance.

1. Introduction

In the domains of automobile, maritime, and aerospace engineering, the pursuit of lightweight metal materials is a persistent scholarly focus. Offering significant weight savings contributing to fuel efficiency [1–3], Fe–Mn–Al–C steel emerges as a material of choice compared with conventional Al alloys [4], Ti alloys [5], dual-phase steel [6] and stainless steel [7], etc, exhibiting superior noise, impact damping capabilities [8] and high strength-ductile in lightweight applications [9–18]. Based on the constituent phases, these lightweight steels are categorized into three types, i.e., ferritic steels, austenitic steels, and dual-phase ferritic-austenitic steels [19]. The austenitic variant of this steel displays commendable mechanical properties, including a tensile strength of over 800 MPa, ductility with elongations exceeding 10%, and a modest density range of 6.8–7.4 g/cm^3 [19]. Nevertheless, while the base yield strength of these steels typically spans 300–800 MPa [19–21], it is important to note that higher yield strengths can be achieved through subsequent aging treatments or by the incorporation of additional alloying elements, as demonstrated in Refs. [19,22,23].

Extensive research into the Fe–Mn–Al–C quaternary system has elucidated that adding Al element can not only efficaciously reduce the density of steel but also strengthen it through the genesis of nanoscale

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https://doi.org/10.1016/j.jmrt.2024.07.188
Received 8 May 2024; Received in revised form 23 July 2024; Accepted 24 July 2024
Available online 27 July 2024
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κ-carbides. Incrementing Al content in Fe–Mn–Al–C-based steels typically yields formidable specific strengths, attributing to a synergistic effect of diminished density and augmented strain hardening [24]. Each 1 wt% increment of Al can cause a 1.3% density reduction [19]. Additionally, Al catalyzes the formation of the κ-carbides, which manifests as both intragranular κ and intergranular κ*-carbides, with the former playing a pivotal role in strengthening austenitic low-density steels [20,25,26]. At the atomic scale within γ-austenite, a mix of Fe, Mn, and Al atoms randomly populate a face-centred cubic lattice. When the preferential occupation of lattice vertex points by Al and the centre of the face by Fe and Mn, an ordered γ-austenite phase with $L1_2$ structure is formed, sometimes referred to as a short-range ordered phase [27]. When C takes over octahedral interstices, the body centre (1/2, 1/2, 1/2), κ-carbides are formed [28]. κ-carbide characterized by a perovskite crystal structure, also known as an $E2_1$ structure, has an ideal stoichiometric of $(Fe, Mn)_3AlC$ [29]. A same structure as the κ-carbides, but partial octahedral interstices position without carbon occupancy is called metastable $(Fe, Mn)_3AlC_x (x < 1)$ phase formed by spinodal decomposition, known as κ′ or $L'1_2$ phase [27,30]. Y. F. An et al. [11] investigated the precipitation and transformation pathways of the coherent nano κ-carbides in austenitic low-density steels with adding Ni, revealing that nano-sized κ′-carbides significantly bolster yield strength. M. J. Yao et al. [21] have posited that ductility remains largely unimpaired due to the coherence between κ-carbides and austenite and the shear ability of κ-carbides by dislocation. References [10,31,32] show that adding Cr mitigates intergranular coarse κ*-carbide precipitation in γ-austenite, which is beneficial for elongation [33–35]. The effect of Ni and Cr elements in controlling the κ-carbides remain largely uncharted.

G. Frommeyer et al. [36]discussed high-strength austenite lightweight steels with a general composition of Fe-(18–28)Mn-(9–12)Al-(0.7–1.2)C steel, distributed by nanoscale κ′-carbide and α-Fe(Al, Mn) ferrite. They delivered a harmonious balance of low density and robust mechanical properties, including high strength and significant ductility. The main deformation mechanism of these steels is the shear-zone-induced plasticity (SIP) effect. Z. Q. Wu et al. [37] investigated the microstructure evolution and deformation mechanism of Fe–12Mn–8Al-0.8C steel, revealing a gradual transition from Taylor lattice to highly dense dislocation walls and microstrips is observed in austenite related to changes in the substructure of such dislocations as the strain increases. The deformation behavior and strengthening mechanism of Cr and Ni alloyed Fe–Mn–Al–C steels have not been studied deeply.

Hence, this work endeavors to prepare low-density steel with a comprehensive excellent property and a density below 6.8 g/cm^3 with exemplary attributes by alloying with 9 wt% Al. Through rigorous EBSD, TEM, and computational analyses, this research comprehensively scrutinizes the microstructural and mechanical properties of Fe–Mn–Al–C steel by Ni and Cr alloying, elucidating the growth mechanisms of nanoscale κ-carbides and examining their strain hardening and deformation behaviors under different cumulative strains. Consequent discussions will dissect the deformation and strengthening mechanics of Fe–Mn–Al–C steel according to the experimental results and the theoretical models, particularly in the context of Ni and Cr alloying, aiming to unravel the mechanisms underpinning their contribution to steel strengthening and ductility effect. This research is a significant contribution to the development of Fe–Mn–Al–C-xNi-yCr austenitic lightweight steels to become more competitive among the widespread lightweight structural materials.

2. Materials and experimental details

Four ingots with nominal compositions of Fe–20Mn–9Al-1.5C-xNi-yCr (x = 0, 2 wt%, y = 0, 3, 5 wt%) were prepared by vacuum induction melting under a high-purity Ar atmosphere. The measured compositions of these as-cast ingots are shown in Table 1. The as-cast ingots were homogenized at 1200 °C for 2 h. The initial and finishing rolling temperature is 1040–1060 °C and 950–970 °C, respectively. The ingots were hot-rolled (HR) to 4 mm-thick sheets with 95% thickness reduction. The plate was online ultrafast cooled (UFCed) to ambient temperature with a cooling rate of ~40 °C/s, based on the efficient-use-water ultrafast-cooled device (ADCOS-PM) developed in-house by our laboratory.

The tensile specimens were cut by electrical discharge machining (EDM) along the rolling direction (RD) of the steel plate with a gauge length of 25 mm, a thickness of 3 mm and a width of 7 mm. Tensile tests were performed at a crosshead speed of 3 mm/min in a Gotech electronic universal testing machine, where a laser extensometer was used to monitor the tensile strain. At least three tensile specimens of each steel were deformed to fracture to ensure reproducibility. To reveal the microstructural evolution during straining, interrupted tensile tests to engineering strains of 3% (or 5%) and 10% were performed.

The microstructure was analyzed by scanning electron microscopy (SEM), electron back-scatter diffraction (EBSD) and transmission electron microscopy (TEM). The possible precipitation structures and their energies for the Fe–20Mn–9Al-1.5C steel were calculated by the Materials Project database (https://legacy.materialsproject.org/). The Materials Project database employs density functional theory (DFT) calculations using the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) functional [38]. EBSD was performed in a Zeiss Ultra55 scanning electron microscope, and specimens were prepared by vibration polishing with 50 nm colloidal silica for 2.5–24 h in a Buehler-VibroMet 2. TEM was performed on a FEI Tecnai G2 F20 instrument operating at 200 kV. The TEM specimens were cut by EDM with a thickness of 1 mm and then first ground into foils with a thickness of <80 μm. Subsequently, the foils were further thinned using Struers TenuPol-3 electrochemical double-jet equipment, which was formed into an electrolyte of 10 vol% perchloric acid and 90 vol% ethanol at a voltage of 20 V and a temperature of −30 °C.

3. Result

3.1. Microstructure

Fig. ($1a_1$-$1d_1$) and Fig. ($1a_2$-$1d_2$) are EBSD inverse pole figures (IPFs) and EBSD phase maps showing the microstructure of the 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels, respectively. Fig. 2 is an X-ray diffraction (XRD) analysis of the 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steel. Both EBSD and XRD analyses reveal that 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels are fully recrystallized, and the microstructures are all composed of a single equiaxed face-centred cubic (FCC) structure γ phase consisting of a high fraction of Σ3 twin boundaries, while no B2 phase with body-centred cubic (BCC) structure was found. Excluding twin boundaries, the average grain sizes of hot rolled 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels are

Table 1
Measured chemical compositions (wt.%) of the four Fe–20Mn–9Al-1.5C-xNi-yCr (x = 0, 2 wt%, y = 0, 3, 5 wt%) steels.

Steels	Chemical composition, wt.%											
	C	Si	Mn	P	S	Al	Mo	Ni	Cr	Ti	Nb	Fe
3Cr	1.31	0.36	20.1	0.011	0.002	8.21	0.51	0.18	2.88	0.01	0.03	Bal.
2Ni	1.23	0.35	21.0	0.010	0.001	9.35	0.57	2.05	0.05	0.01	0.03	Bal.
2Ni3Cr	1.41	0.35	20.3	0.009	0.002	8.63	0.53	2.11	2.89	0.01	0.03	Bal.
2Ni5Cr	1.34	0.37	20.8	0.010	0.002	9.09	0.56	2.08	4.90	0.01	0.03	Bal.

nearly equivalent, being 7.63 μm, 8.95 μm, 8.57 μm and 7.60 μm, respectively. When they are included, the average grain sizes are 3.51 μm, 3.73 μm, 3.72 μm and 2.2 μm.

Fig. (1e–1h) are TEM bright-field images showing similar results with EBSD IPF images, in which the microstructures are all γ austenite phase. For 3Cr and 2Ni5Cr steels which have higher Cr content, Fig. 1i and l are dark-field (DF) TEM images revealing nanosized long range ordering (LRO) domains homogeneously distributed in the γ matrix. The selected area diffraction patterns (SADPs) taken along the [011] γ zone axis of the 3Cr and 2Ni5Cr steels are shown in Fig. 1m and p, respectively. Very faint superlattice reflections can be seen at the 1/2 $\{0–22\}_\gamma$ or/and 1/2 $\{200\}_\gamma$ positions, and satellite diffraction spots around the

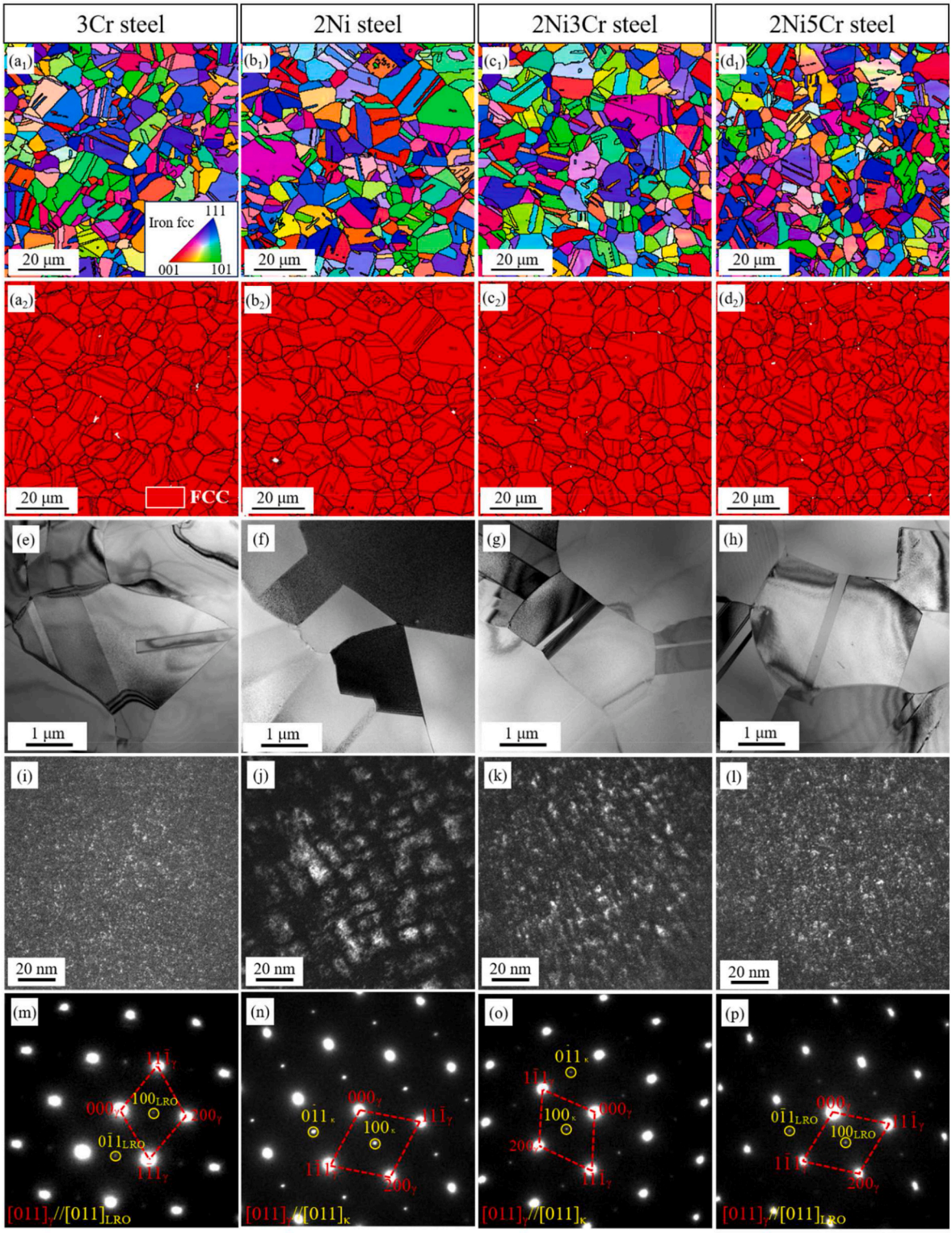


Fig. 1. Microstructures of the 3Cr steel, 2Ni steel, 2Ni3Cr steel and 2Ni5Cr steel. (a_1, b_1, c_1, d_1) EBSD IPF maps. (a_2, b_2, c_2, d_2) EBSD phase maps. (e, f, g, h) TEM bright-field images. (i, j, k, l) TEM dark-field images taken from the {110} LRO or {110} κ reflections. (m, n, o, p) Selected area diffraction patterns recorded along the [001] γ zone axis: superlattice intensities from the κ-carbides and LRO domains and the primary γ diffraction spots.

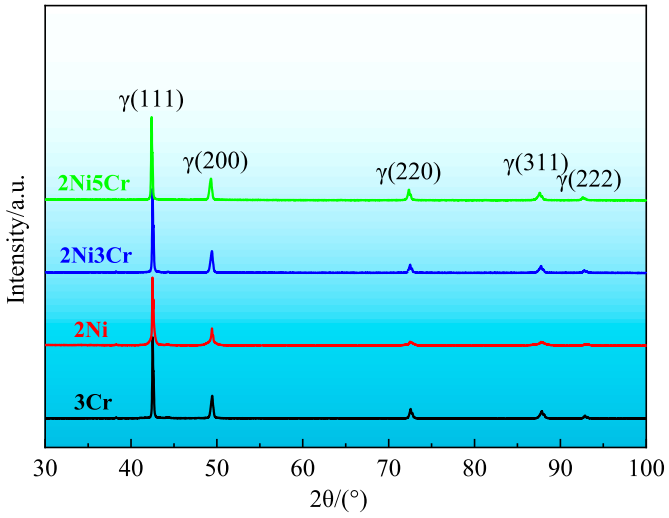


Fig. 2. X-ray diffraction analysis of the 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steel.

primary γ diffraction spots cannot be found, indicating the formation of LRO domains in the online UFCed 3Cr and 2Ni5Cr steels. No satellite diffraction spots around γ diffraction spots and high carbon content in Fe–Mn–Al–C lightweight steels illustrate the formation of the κ-carbides/LRO domains rather than κ′-carbides [14,39]. For 2Ni and 2Ni3Cr steels, which have higher Ni content, Fig. 1j and k reveal orthorhombic κ-carbides distributed in the γ matrix. The 2Ni steel has the largest size of κ-carbides with an average size of 10 nm. Fig. 1j and k demonstrate that the ordered structures of 2Ni and 2Ni3Cr steel are κ-carbides with cuboidal morphologies [21,40]. The low intensity of the superlattice diffraction indicates that the volume fraction of the κ-carbides is very low. However, the thickness of the sheet is uncertain, so a quantitative estimate of the volume fraction cannot be given from the DF.

Figs. (3a, 3c, 3e and 3g) are atomic resolution HRTEM images along the [011]γ zone axis showing the microstructures of the LRO domains in 3Cr and 2Ni5Cr steels (Fig. 3a and g) and the κ-carbides in 2Ni and 2Ni3Cr steel (Fig. 3c and e), respectively. The inset patterns in Fig. 3a, c, e and g are fast Fourier transform (FFT) patterns. To more clearly and directly reveal these LRO domains in the HRTEM imaging, inverse FFT (IFFT) images were obtained using only superlattice reflections, as shown in Fig. 3b, d, 3f and 3h. As shown in Fig. 3b and h, the microstructure of LRO domains is ellipsoidal, and the spatially localized LRO domains have an interconnected morphology, which is often observed in spinodal decomposition. The average sizes of the LRO domains in 3Cr steel and 2Ni5Cr steel measured using IFFT images are approximately 1.75 nm and 2.14 nm, respectively. The average sizes of these LRO domains and κ-carbides determined by measuring more than 200 domains by Image-Pro Plus software are plotted in Fig. 3i. Their average sizes decrease from ~10.69 nm through ~5.46 nm and ~2.14 nm–~1.75 nm for the 2Ni steel, 2Ni3Cr steel, 2Ni5Cr steel and 3Cr steel.

In Fig. 3, it can be seen that the size of κ-carbide decreases with increasing content of Cr elements. We hypothesize that the addition of Cr will negatively impact the formation of κ-carbide/LRO domains, whereas Ni will promote the generation of κ-carbides. This idea will be demonstrated in Section 4.1 and was also confirmed in the text [24,41].

3.2. Tensile response

Fig. 4a shows the engineering stress-strain curves of UFCed 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels with equivalent grain sizes. This indicates that the total elongation of steel with a higher content of Cr (i.e., 2Ni5Cr steel) is higher than others (i.e., 2Ni and 2Ni3Cr steel), and the highest total elongation was found in the 2Ni5Cr steel with an increment of ~16% for comparison with no Cr steel (i.e., 2Ni steel). The yield strength of the steel improves with the addition of Ni. Compared to 3Cr steel, steel with a higher Ni content (i.e., 2Ni3Cr steel) has a yield strength increase of ~141 MPa.

These four steels have similar grain sizes, but these steels show an increase in elongation with an increase in Cr content and an increase in yield strength with an increase in Ni, indicating that a strength-ductility combination can be achieved by alloying with Ni and Cr. The detailed reasons for this will be discussed in Discussions 4.2 and 4.3. The tensile properties of these four steels are listed in Table 2, from which it can be seen that in 2Ni, 2Ni3Cr and 2Ni5Cr steels, the yield strength decreases from 1202 MPa to 1071 MPa with the addition of Cr, and the tensile strength decreases from 1283 MPa to 1187 MPa. However, the addition of Cr in 2Ni, 2Ni3Cr and 2Ni5Cr steels increases the total elongation from 34.1% to 54.5%, and the total elongation of 3Cr steels without added Ni is up to 54.4%. 3Cr steels have a strong plasticity of up to

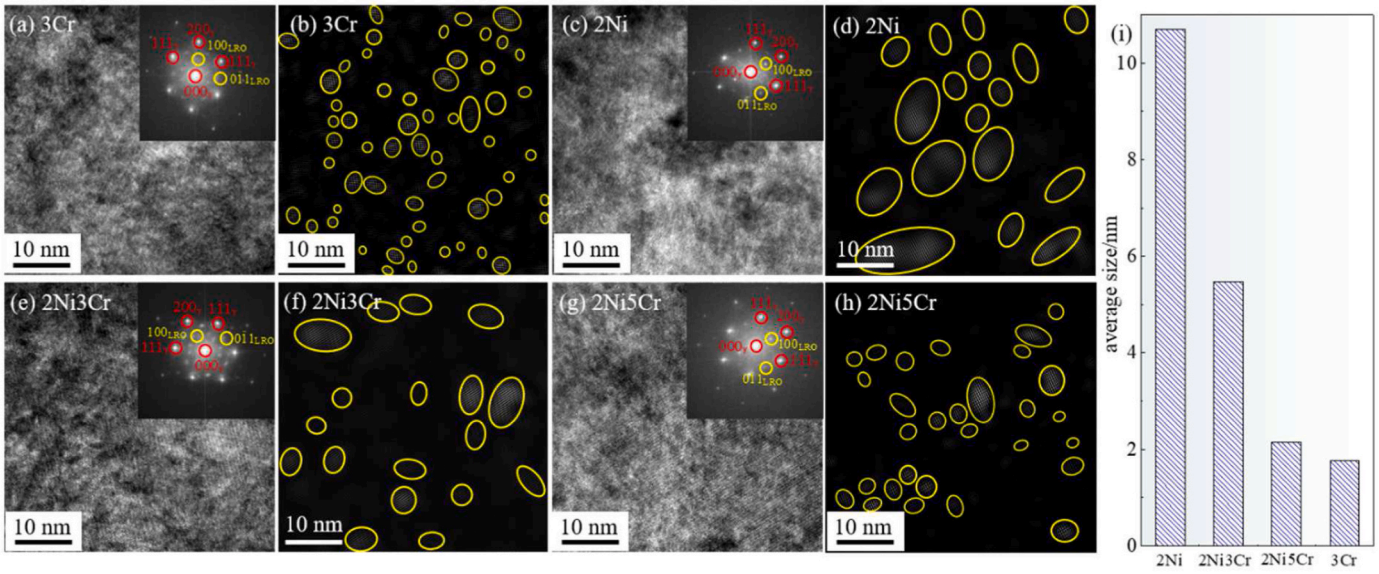


Fig. 3. High-resolution TEM images (HRTEM) imaging of LRO domains and κ-carbides in Fe–20Mn–9Al-1.5C-xNi-yCr steel: (a, c, e, g) HRTEM imaging along the [011]γ zone axis of 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steel, respectively; (b, d, f, h) IFFT image of Fig. 3a, c, 3e and 3g using {011} and {100} superlattice reflections of LRO domains; (i) Average size of the LRO domains or κ-carbides.

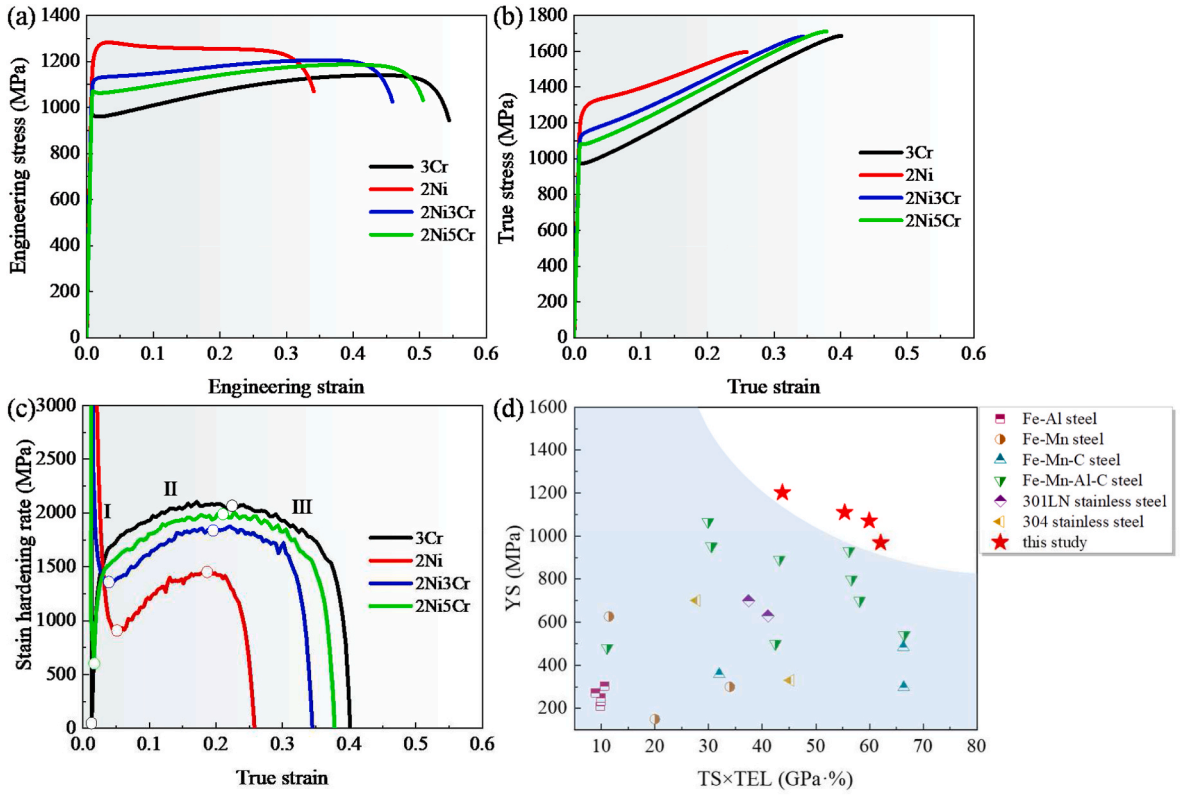


Fig. 4. Tensile properties of 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steel. (a) Engineering stress–strain curves. (b) True stress–strain curves. (c) Strain hardening curves. (d) TS × TEL versus tensile strength. The experimental data for Fe-Al [42], Fe-Mn [43,44], Fe–Mn–C [45,46,47], Fe–Mn–Al–C [14,36,45,46,48] and stainless steels [49,50] are compared with those for the present Ni&Cr-alloyed steels, and the raw data for the tensile test are given in Table 2.

Table 2
Tensile results for UFCed 3Cr, 2Ni, 2Ni3Cr, 2Ni5Cr steel and reference steels.

	YS, MPa	TS, MPa	TEL, %	TS × TEL, GPa·%	Reference
Fe–8Al	210	281	35	9.83	[42]
Fe–9Al	233	302	33	9.97	[42]
Fe–10Al	249	311	32	9.95	[42]
Fe–11Al	272	331	27	8.94	[42]
Fe–12Al	304	366	29	10.61	[42]
Fe–10Mn	627	961	12	11.44	[43]
Fe–24Mn	300	809	42	33.98	[44]
Fe–30Mn	150	500	40	20.0	[44]
Fe–14Mn-1.05C	358	653	49	31.99	[45]
Fe–18Mn-0.6C	484	1105	60	66.30	[46]
Fe–22Mn-0.5C	298	897	74	66.38	[47]
Fe–14Mn–3Al-1.05C	480	554	20	11.08	[45]
Fe–18Mn–3Al-0.6C	499	849	50	42.45	[46]
Fe–21Mn–6Al–1C–3Si	930	1040	54	56.16	[14]
Fe–28Mn–12Al–1C	700	881	66	58.14	[36]
(TRIPLEX steel)	798	885	64	56.64	
	891	939	46	43.19	
	952	985	31	30.54	
	1065	1068	28	29.90	
Fe–30Mn–8Al–1.2C (TRIPLEX steel)	540	898	74	66.45	[48]
Fe–17Cr-6.5Ni-1.2Mn	700	1070	35	37.45	[49]
(301LN stainless steel)	630	1027	40	41.08	[49]
Fe–18Cr–9Ni-1.8Mn (304	701	805.5	34.5	27.79	[50]
stainless steel)	330	644.6	70.2	45.25	[50]
Fe–20Mn–9Al-1.5C–3Cr	969	1141	54.4	62.07	This study
Fe–20Mn–9Al-1.5C–2Ni	1202	1283	34.1	43.75	
Fe–20Mn–9Al-1.5C–2Ni3Cr	1110	1206	45.9	55.36	
Fe–20Mn–9Al-1.5C–2Ni5Cr	1071	1187	50.5	59.94	

62.07 GPa·%. Compared with 2Ni and 2Ni3Cr steel, the yield and tensile strengths of 2Ni steels are 1202 MPa and 1283 MPa, respectively, and those of the 3Cr steels with no added Ni, are even lower, at 969 MPa and 1141 MPa, respectively. Regarding the mechanical properties of the present Ni–Cr-alloyed steel, other single-phase and two-phase light-weight steels, conventional TWIP steels and stainless steels [42–47, 49–51] is shown in Fig. 4d and Table 2. This shows that the superior combination of strength and ductility of Fe–20Mn–9Al-1.5C-xNi-yCr steels distinguishes them from other steels, especially in terms of a significant increase in yield strength for the same GPa·% or total elongation. Obviously, for the present Ni and Cr alloy steels with equivalent grain sizes, the increase in strength with the addition of Cr and Ni should be associated with the formation of LRO domains or κ-carbides.

The true stress−strain curves and work-hardening curves of 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steel are shown in Fig. 4b and c, respectively. There are two types of work-hardening curves between the work-hardening curves of steels. One type includes the black and green curves representing 3Cr and 2Ni5Cr steel, respectively, and the other includes the blue and red curves representing 2Ni and 2Ni3Cr steels, respectively. The hollow dots in Fig. 4c indicate that there are three strain hardening stages (Stages I, II, and III) as the true strain increases, as observed for the four steels. During Stage I, the strain hardening rate of the four steels decreases sharply with increasing strain. During Stage II, the strain hardening rate tends to increase with increasing strain, and Stage II is prolonged compared among the 2Ni, 2Ni3Cr and 2Ni5Cr steels with increasing Cr content. The strain hardening rate of the four steels in Stage III decreases, and the work hardening rate of 2Ni5Cr is higher than that of the 2Ni and 2Ni3Cr steels with increasing Cr content in the process of Stage III. During Stage III, 2Ni3Cr steel has a higher work hardening rate than 2Ni steel. Regarding the Ni comment, comparing the 3Cr and 2Ni3Cr steels, as the Ni content increases, Stage II and Stage III are shortened and lowered. These changes in the strain hardening

behavior should be responsible for the significant enhancement of tensile strength and ductility with the addition of Ni and Cr.

In order to characterize the fracture of the steels, local SEM images of the fibre zone for the UFCed 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels are shown in (Figs. 5a–d), respectively, and the insets of Fig. 5 show the macroscopic morphology of the fracture surfaces. A large number of dimples can be observed in the fibre zone of the four steels. This indicates that they all fracture in a ductile mode.

3.3. Microstructural evolution upon deformation

To further study substructure evolution at different stages of plastic deformation, TEM specimens were prepared from interrupted test specimens and tensile specimens tested to fracture. Fig. 6 represents the evolutionary characteristics of the substructural deformation of 3Cr steel at different engineering strain stages. The bright field TEM images were taken along the $[011]_\gamma$ zone axis of the FCC matrix phase, and the inset shows a selected area electron diffraction pattern. As shown in Fig. 6a–c, when the strain reaches 3%, dislocation slip is the main feature, while the [011](1-11) slip system in the grain has a Schmid factor of 0.52. The dislocation slip has a strong planarity, such as dislocation multipoles localized within {111} slip bands (Fig. 6a), dislocations constrained within planar (1-11) slip bands and restricted by the planar slip bands (Fig. 6b) and the interaction of (1-11) and (11-1) slip bands from different slip systems (Fig. 6c), and has also been observed in other FCC alloys with similar phenomena [52].

With strain accumulating to 10%, dislocation slip is also the main feature, while the slip bands are mainly (1-11) slip bands. It is also illustrated that the space between the slip bands becomes narrower, the number of slip bands increases, and the dislocation density increases, as shown in Fig. 6d. There are also more slip systems operated, and planar slip bands from different slip systems appear as (1-11) and (11-1) slip surfaces (Fig. 6e and f). They interrupt each other. (1-11) and (11-1) slip bands, which carry high-density dislocation arrays (HDDA); HDDA is restricted by the (1-11) and (11-1) planar slip bands (Fig. 6e).

When the strain reaches 54% (LD), the microstructural evolution is shown in Fig. 6g–l. High-density dislocations and very fine slip bands can be observed (Fig. 6g–h), and a dozen nanometre-width deformation twins are seen in the bright-field image (Fig. 6i), the selected electron diffraction pattern (in the inset of Fig. 6i) and TEM high-resolution images (Fig. 6j). Dozens of nanometre-width stacking faults can be observed in the bright-field image (Fig. 6k), the selected electron diffraction pattern (in the inset of Fig. 6k) and the dark-field image (Fig. 6l). Since the deformation twins appear only in the vicinity of the fracture surfaces, their contribution to the overall ductility and strain-hardening capacity is expected to be very limited. This implies that 3Cr steels are dominated by HDDA and dislocation slip during deformation, and their strain hardenability is mainly contributed by HDDA densification and dynamic slip band refinement (DSBR).

Fig. 7 reveals the evolution of the deformation substructure microstructures at different strains of 2Ni steel. The deformation characteristics are primarily dominated by planar slip. With strain increasing to 5%, slip bands are only observed, as shown in Fig. 7a and b. Deformation characteristics reveal the appearance of (1-11) slip bands (Fig. 7a), and it is also observed that the deformation characteristics are still slip bands on the (−111) slip plane intercepted by the slip bands on the (1-11) slip plane (Fig. 7b). As the strain accumulates to 10%, the microstructural evolution can be seen in Fig. 7c–g and is characterized by the multiplication of planar slip bands as well as the gradual refinement space between the slip band (Fig. 7c–f) and the (200) slip band, forming three groups of slip systems: [011](1-11), [011](−111) and [011](200) (Fig. 7g). When the strain begins to fracture, the deformation microstructure shows high-density dislocations (Fig. 7h) and very fine slip bands (Fig. 7i). Stack faults were also found in bright-field images (Fig. 7j), selected electron diffraction patterns (in the inset of Fig. 7j), dark-field images (Fig. 7k) of Fig. 7j and TEM high-resolution images (Fig. 7l) which is the circled portion of Fig. 7k, but no deformation twins were detected. As shown in Fig. 8a and b, the near-fracture state of 2Ni

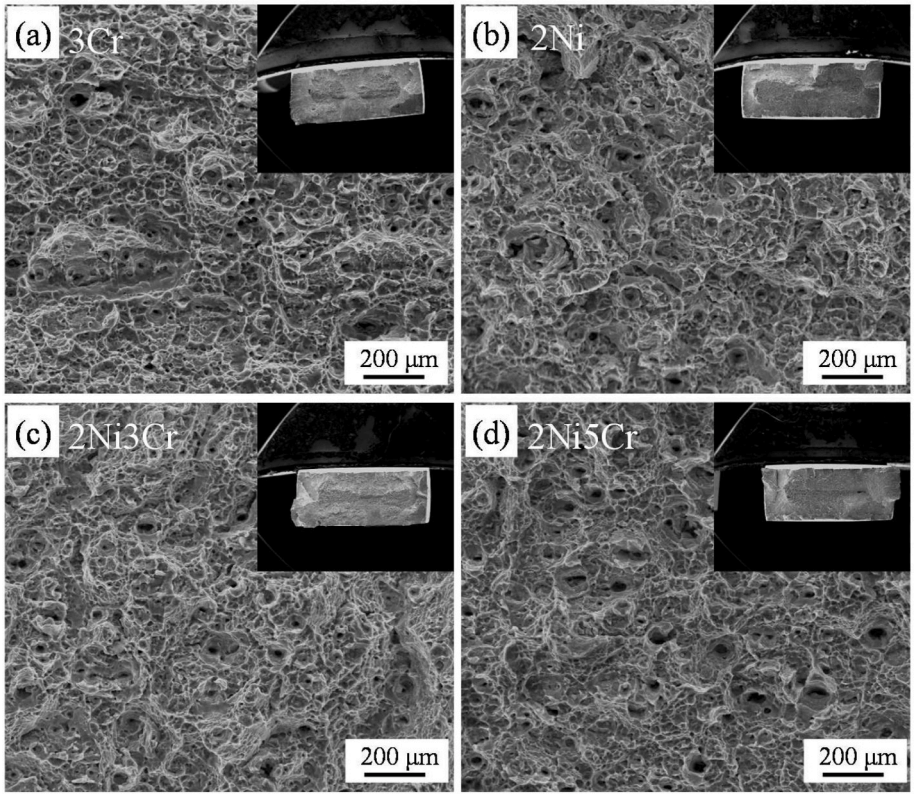


Fig. 5. SEM images of the fracture surfaces of (a) 3Cr, (b) 2Ni, (c) 2Ni3Cr and (d) 2Ni5Cr steel.

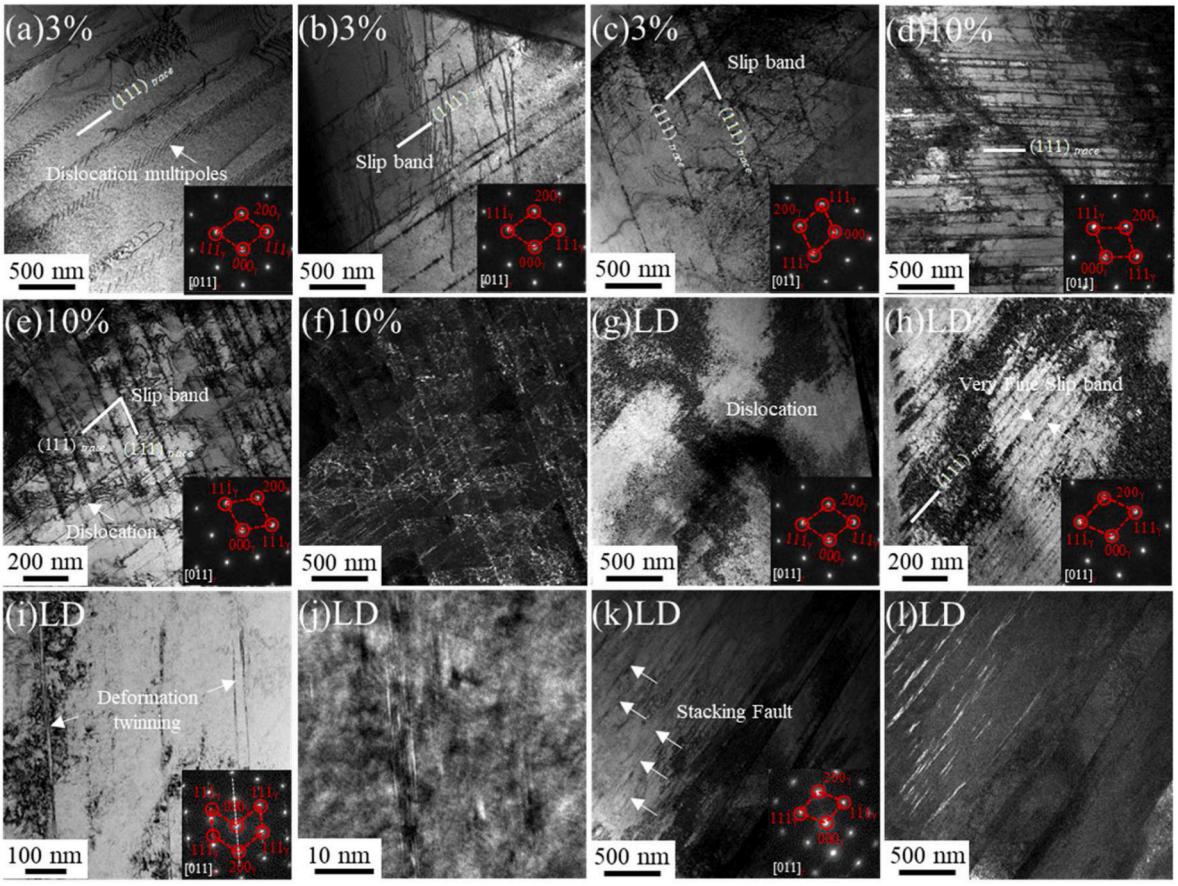


Fig. 6. TEM images of the microstructural evolution upon deformation of 3Cr steel at various engineering strains: (a–c) 3%, (d–f) 10%, (g–l) close to final failure (LD).

steel was also found to exhibit dislocation plane slip after deformation under loading, and the plane slip penetrated the grain and cut through the κ-carbides. The absence of deformation twins throughout the entire plastic deformation of the 2Ni steel. This implies that the strain hardenability of the steels mainly benefits from the DSBR effect.

4. Discussion

4.1. Formation of an $E2_1$-type ordered structure promoted by Cr or Ni alloying

Mechanical properties are closely related to the evolution of the microstructure. In this study, TEM characterization of Fe–20Mn-9A1-1.5C-xNi-yCr lightweight steel austenite is shown in Fig. 1. The formation of κ-carbides in austenite is in accordance with the nucleation and growth mechanism, and the different degrees of the ordered phases and the ordered domains of LRO and κ-carbides lead to different yield strengths [20,53]. Previous studies have shown that the addition of Al can promote the formation of ordered phases [19,20,53]. In this study, the steel under investigation contains not only the elements Fe, Mn, Al, and C but also Ni and Cr, which are added to influence the formation process of the κ-carbides. As depicted in Fig. 1, these two elements indeed affect the phase transition process of the κ-carbides.

For studying phase transformation and thermodynamic properties, formation energy is a crucial parameter as it relates to the stability and equilibrium of the κ-carbides. This study analyzed the effect of Ni and Cr on LRO and κ-carbides in Fe–20Mn-9A1-1.5C steel by first-principle calculations for the Fe–20Mn-9A1-1.5C lightweight steel studied. The possible structures and their energies for the Fe–20Mn–9Al-1.5C steel were calculated by the Materials Project database (https://legacy.materialsproject.org/). The Materials Project database employs density functional theory (DFT) calculations using the generalized gradient approximation (GGA) with the Perdew-Burke-Ernzerhof (PBE) functional [38]. Fig. 9 shows the crystal structures of (a) disordered γ, (b) ordered γ, (c) κ-carbide, and (d) $AlNi_3C$ (one of the Ni-rich κ-carbide). In Fe–20Mn-9A1-1.5C-xNi-yCr lightweight austenitic steels, the Fe, Mn, and Al atoms randomly occupy the vertex and face-centre positions of the face-centred cubic lattice (Fig. 9a). When the Al element occupies the position of the vertex and the Fe and Mn elements occupy the face centre positions of the cubic lattice, the ordered γ-austenite phase is formed (Fig. 9b). When elemental C gradually occupies the body-centred (1/2, 1/2, 1/2) octahedral interstitials, κ-carbides are formed. κ-carbides are perovskite crystal structures, also known as $E2_1$ structures [29]. The ideal stoichiometry of κ-carbides is $(Fe, Mn)_3AlC$ [29]. The results of the first-principle calculations for the possible formation of κ-carbides structure are shown in Table 3.

With the addition of the Cr element into the κ-carbides, the formation energy of the κ-carbides increases, requiring greater energy for phase transition, thus making it more difficult for κ-carbides nucleation and growth. Thus, the size of the κ-carbides decreases. Conversely, with the addition of Ni element into the κ-carbides, the formation energy of the κ-carbides decreases, requiring less energy for phase transition, thus making it easier for κ-carbides nucleation and growth. In the calculated results, as shown in Table 3, the addition of Cr into the steels has the potential to form the carbides with $Cr_{21}(FeC_3)_2$ structure in the γ-matrix. According to the calculated formation energy of the $Cr_{21}(FeC_3)_2$ structure, which is −0.049 eV/atom, being negative indicates that the $Cr_{21}(FeC_3)_2$ structure of the carbides can form in nature. However, compared to the most commonly formed $(Fe, Mn)_3AlC$ structure, it is higher by 0.111 eV/atom, increasing the nucleation energy barrier.

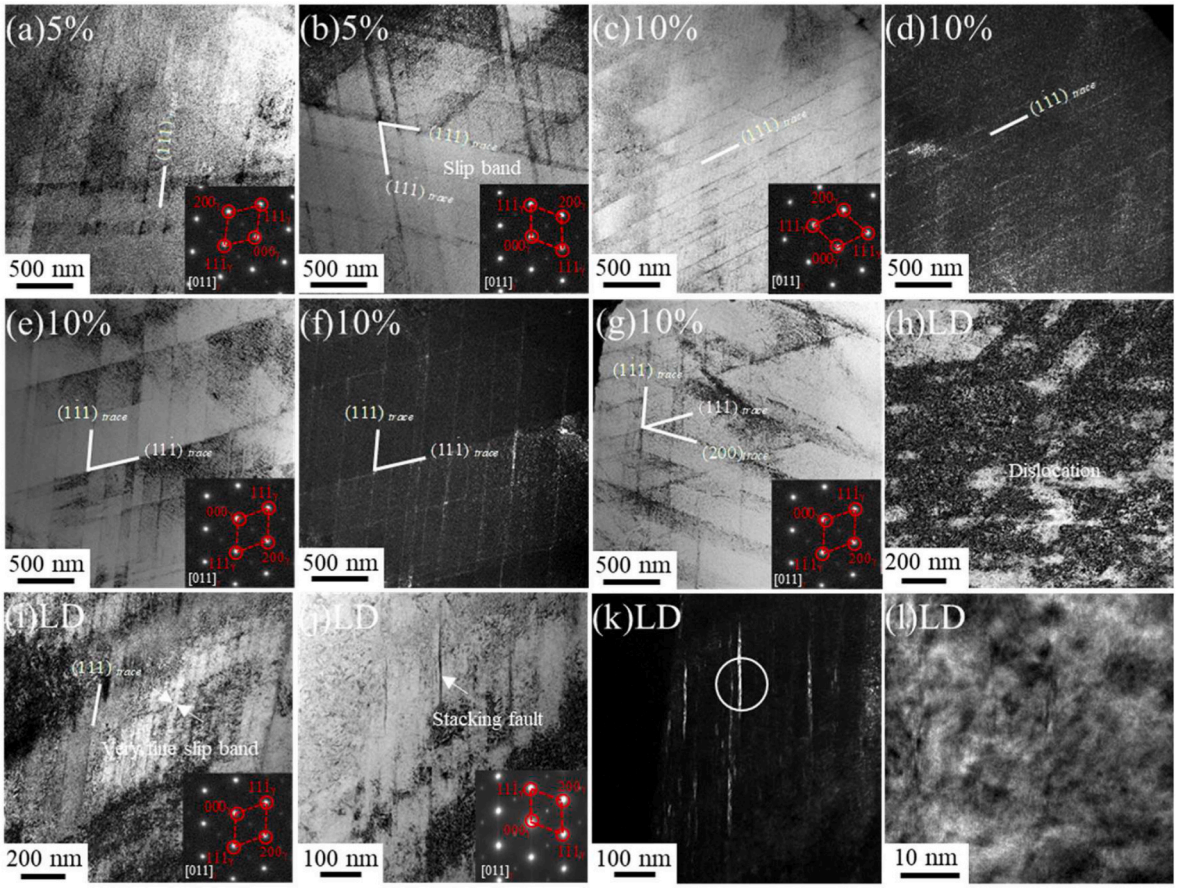


Fig. 7. TEM images of the microstructural evolution upon deformation of 2Ni steel at various engineering strains: (a–b) 5%, (c–g) 10%, (h–l) close to final failure (LD).

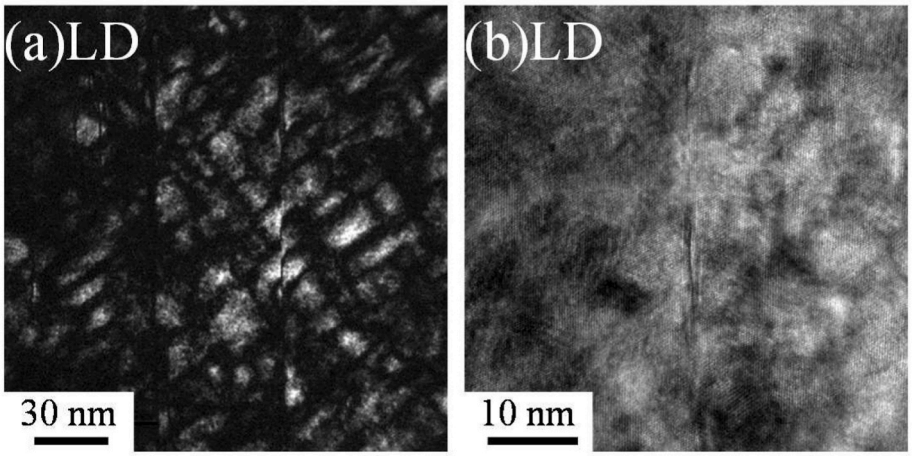


Fig. 8. TEM images of 2Ni steel close to final failure: (a) dark field image of κ-carbides, (b) HRTEM of κ-carbides.

Therefore, the addition of Cr into the Fe–Mn–Al–C matrix plays a suppressing role in the generation of the carbides. And Cr atoms take the place of Al atoms to form $(Fe, Mn)_3CrC$ [54]. This also increases the formation energy of κ-carbides, as well as the interfacial energies and elastic strain energies between κ-carbides and the austenite matrix, causing an increase in the nucleation energy barrier of κ-carbides. This phenomenon can also be found in 3D APT characterization [14] and first-principles calculations [24]. Factors such as the increase in the nucleation energy barrier are the main reasons for the decrease in the size of κ-carbides in Cr-containing Fe–Mn–Al–C lightweight steels. For 2Ni steels, since Ni has an atomic radius of 1.24°A, Fe atoms have a radius of 1.27°A, and Mn atoms have a radius of 1.32°A. They have equivalent atomic radii, and Ni atoms preferentially displace Fe/Mn atoms to form Ni-rich κ-carbides with $AlNi_3C$ structure (with a formation energy of −0.181 eV/atom). The formation energy of Ni-rich κ-carbides with $AlNi_3C$ structure is lower than that of ideal κ-carbides. The addition of Ni atoms in κ-carbide reduces the formation energy of κ-carbide, lowers the critical formal energy of κ-carbide, and promotes the formation of κ-carbide. The reduction in the nucleation energy barrier is the main reason for increasing the size of κ-carbide in Ni-containing

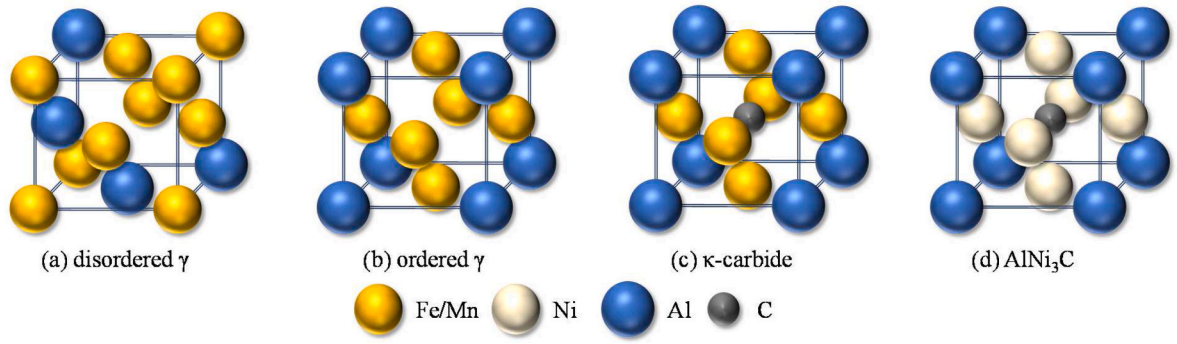


Fig. 9. Crystal structures of (a) disordered γ, (b) ordered γ, (c) κ-carbide, and (d) $AlNi_3C$ (one of the Ni-rich κ-carbide).

Table 3
First-principle calculations results of formation energy for the possible κ-carbides structure formation.

	possible (κ-) carbides structure	Formation energy (eV/atom)
κ-carbides	$(Fe, Mn)_3AlC$	−0.160
Cr-rich carbides	$Cr_{21}(FeC_3)_2$	−0.049
Ni-rich carbides	Fe_2NiC	0.082
	$AlNi_3C$	−0.181

lightweight steels. The results are consistent with those of the dark field images of Fig. 1i–1l in Section 3.1. Since the nucleation energy of the Fe_2NiC structure is > 0, it is less likely for the Fe_2NiC structure of the κ-carbides to form in nature.

4.2. The contribution increasement in yield strength after Ni–Cr alloying

We can predict that the yield strength of steels in this study is primarily contributed by LRO nanodomains and κ-carbides. The yield strength can be represented as a linear combination of strengthening mechanisms, as illustrated in Equation (1) [14]. The influence of dislocation strengthening in fully recrystallized Fe–Mn–Al–C steel is minimal and is therefore disregarded.

$$\sigma_{ys} = \sigma_0 + \sigma_{ss} + \sigma_{gb} + \sigma_{prep} \tag{1}$$

here, σ_0 can be readily calculated based on the shear stress of dislocation motion(τ_0): $\sigma_0 = 3\tau_0$ [55], and represents the sum of the frictional stress components. According to Honeycombe [56], $\tau_0 = 2 \times 10^{-4}G$, where G is the shear modulus. Given that the G value for Fe–Mn–Al–C steel is 70 GPa, σ_0 here is calculated to be 42 MPa. The Hall-Petch (HP) relationship [57,58], $\sigma_{gb} = K_{HP}d_{eff}^{-0.5}$, is employed to fit the experimental data and to represent the grain boundary strengthening component σ_{gb}. Here, K_{HP} represents the Hall-Petch coefficient for steels in this study (K_{HP} = 446.8 MPa $\mu m^{0.5}$ [59]). d_{eff} represents the grain sizes containing twins for the 3Cr, 2Ni, 2Ni3Cr, and 2Ni5Cr steels as described in Section 3.1, and we obtained the d_{eff} values from EBSD analysis as 3.51, 3.73, 3.72, and 2.2 μm, respectively. As shown in Fig. 1, the effects of Ni and Cr elements on the grain sizes of Fe–Mn–Al–C austenite steels are minimal. Therefore, the calculated strengthening component results for the 3Cr, 2Ni, 2Ni3Cr, and 2Ni5Cr specimens are quite similar, amounting to 238.5, 231.3, 231.7, and 301.2 MPa, respectively. In Equation (1), σ_{ss} represents the solute strengthening component expressed as an empirical formula: σ_{ss} = 279[C]-1.5[Mn]+20.5[Al]+2.9[Ni]+3.9[Cr] [60], where [C], [Mn], [Al], [Ni] and [Cr] represent the weight percentages of these elements dissolved in the matrix. The solute strengthening components for the 3Cr, 2Ni, 2Ni3Cr, and 2Ni5Cr samples are 515.4, 509.5, 557.2 and 554.1 MPa, respectively.

In Equation (1), σ_{prep} represents precipitation strengthening components, which can be expressed in various ways depending on the type of dislocation movement, such as shearing and Orowan bypass mechanisms [61]. In this study, as shown in Fig. 8, moving dislocations shear the precipitates, thus, shear strengthening is the predominant mechanism. The shearing mechanism involves three strengthening processes: ordering strengthening (σ_{order}), coherency strengthening (σ_{coh}), and modulus mismatch strengthening (σ_{mod}). These contributions to the yield strength of Fe–Mn–Al–C steel from each mechanism can be predicted using the following equations:

$$\sigma_{order} = \sum_{r=0}^{n} A_S M 0.81 \frac{\gamma_{APB}}{2b}\left(\frac{3\pi f_i}{8}\right)^{1/2} \tag{2}$$

$$\sigma_{coh} = \sum_{r=0}^{n} A_S M \alpha_\varepsilon (G\varepsilon_c)^{3/2}\left(\frac{r_i f_i}{0.5Gb}\right)^{1/2} \tag{3}$$

$$\sigma_{mod} = \sum_{r=0}^{n} A_S M 0.0055(\Delta G)^{3/2}\left(\frac{2f_i}{G}\right)^{1/2}\left(\frac{r_i}{b}\right)^{(3m/2)-1} \tag{4}$$

Here, A_s represents a factor for the strong obstacle precipitation phase, and κ-carbide is considered a strong obstacle since the precipitation fraction of κ-carbides in Fe–Mn–Al–C steels is significantly higher than that of traditional steels. In this study, we set A_s to 0.64 [59]. Additionally, $M \approx 3.06$ represents the Taylor factor for fcc matrix steels [62]. G is the shear modulus of austenite, and $\Delta G \approx 23.3$ GPa represents the difference in shear modulus between the austenite matrix and κ-carbides [61]. b = 0.264 nm represents the magnitude of the Burgers vector for dislocations in the austenite phase [53]. $\alpha_\varepsilon \approx 2.6$ and $m \approx 0.85$ are constants for fcc metals [61]. Additionally, f_i and r_i respectively represent the radius and fraction of precipitates. ε_c = 0.9% is the lattice parameter mismatch parameter between the austenite matrix and κ-carbides [61]. Finally, $\gamma_{APB} \approx 0.35$ J/m^2 represents the antiphase boundary energy per unit area generated when the ordered κ-carbide is sheared [63].

Here, we assume that the larger of the two factors (σ_{order} and $\sigma_{coh}+\sigma_{mod}$) contributes to the precipitation strengthening [64]. Among these, $\sigma_{coh}+\sigma_{mod}$ peaks when shear dislocations approach the interface between the matrix and the precipitate, while σ_{order} peaks when half of the precipitate is sheared by the dislocation [62]. We calculate σ_{order} and $\sigma_{coh}+\sigma_{mod}$ using the size and volume fraction of nanodomains and κ-carbides, taking the maximum value as σ_{prep}. We ultimately predict the yield strength using size distribution (r_i) and the volume fraction (f_i) of κ-carbides of different sizes, comparing it with the experimental results of tensile tests. The calculated σ_{prep} is shown in Fig. 10, where the red dashed line represents the transition radius, at which the dominant strengthening mechanism shifts from σ_{order} to $\sigma_{coh}+\sigma_{mod}$. For the 3Cr, 2Ni, 2Ni3Cr and 2Ni5Cr steels, the strength contribution of $\sigma_{coh}+\sigma_{mod}$ is dominant. However, in the 3Cr and 2Ni5Cr samples with LRO nanodomains, σ_{order} also contributes to σ_{prep} because some of the nanodomains are smaller than the critical size.

Based on Equation (1), we summarize the strengthening effects for the 3Cr, 2Ni, 2Ni3Cr, and 2Ni5Cr samples. Fig. 11 compares the results

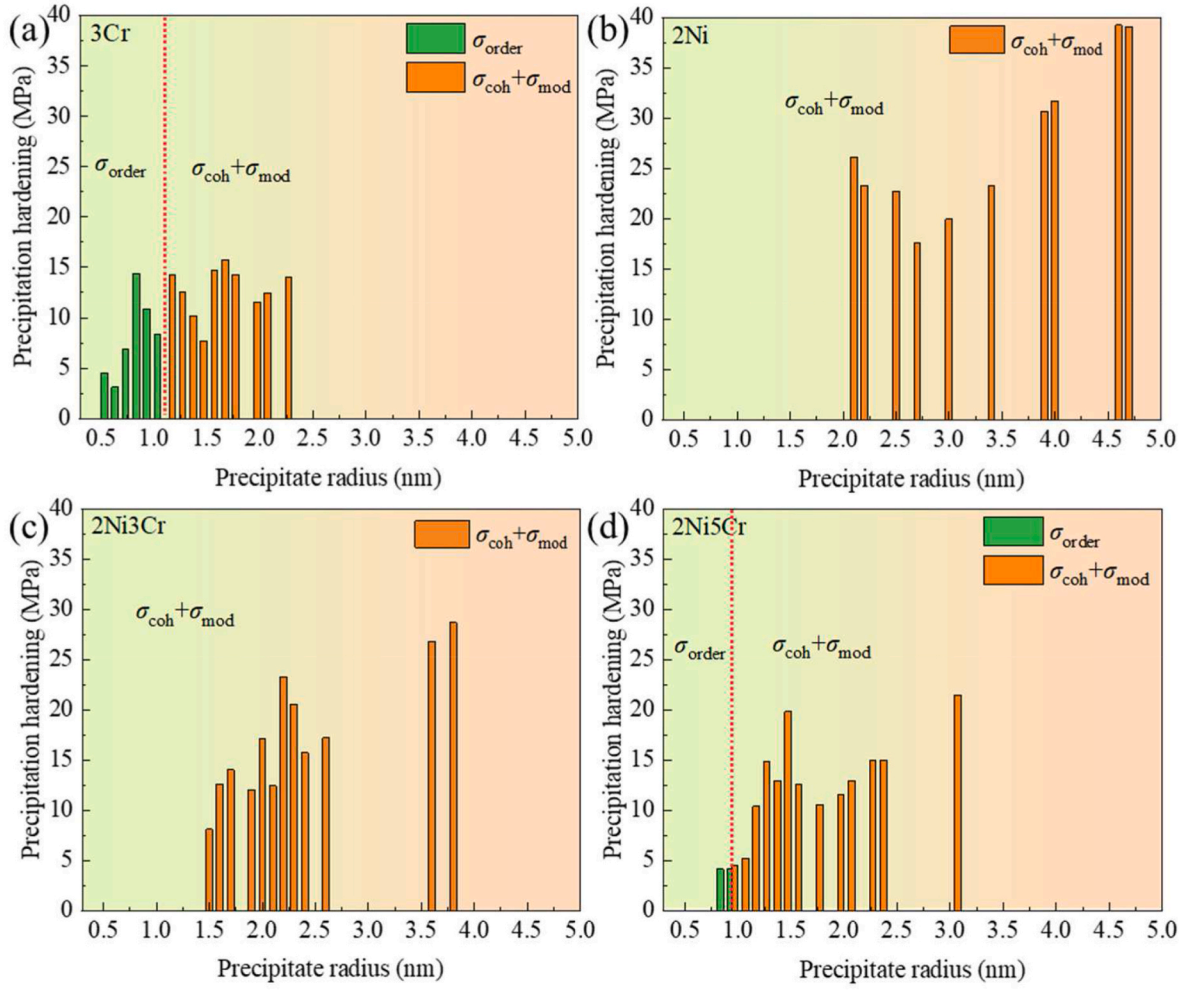


Fig. 10. The precipitation hardening at each precipitate radius according to the size distribution of the precipitates for (a) 3Cr, (b) 2Ni, (c) 2Ni3Cr, and (d) 2Ni5Cr steels.

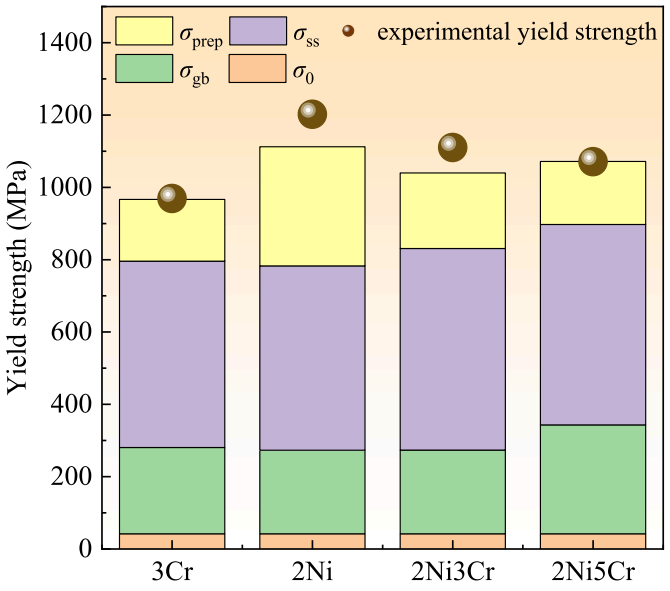


Fig. 11. Schematic of the yield strengths measured by tensile tests and the calculated values of yield strengths produced by the four mechanisms.

with the measured yield strengths, illustrating the contribution to the yield strength of each strengthening mechanism for each sample. There is a correlation between the calculated and experimental values of yield strength and the predicted and measured yield strengths, which are in very good agreement. Solution strengthening is evidently the primary strengthening mechanism for these samples, with the enhancement in yield strength for traditional Fe–Mn–Al–C steels being closely associated with a carbon content as high as 1.5%. Precipitation strengthening also plays a significant role in these samples, with the contribution from Ni and Cr elements synergistically in Fe–Mn–Al–C steel providing an increase in the size of the precipitate phase, leading to a gradual increase in precipitation strengthening as the precipitate size grows.

4.3. *Evolution of deformed substructures during the strain process*

According to the experimental results, the evolution of the deformation-induced microstructure with increasing strain for 2Ni and 3Cr steels is schematically shown in Fig. 12. Fig. 12a–c show the case of 3Cr steel. At the initial stage of plastic deformation, the steel shows planar dislocation activity. For dislocations that initially appear slide in a planar feature, the activity of subsequent dislocations tends to be constrained within the planar slip bands (Fig. 6a–c and Fig. 12a), and the strain hardening rate decreases rapidly, followed by an increase with increasing strain (Figs. 6b and 12a), which is mainly attributed to the larger mean free path of the dislocations. A limited number of dislocation annihilation events at grain boundaries are also expected to lead to

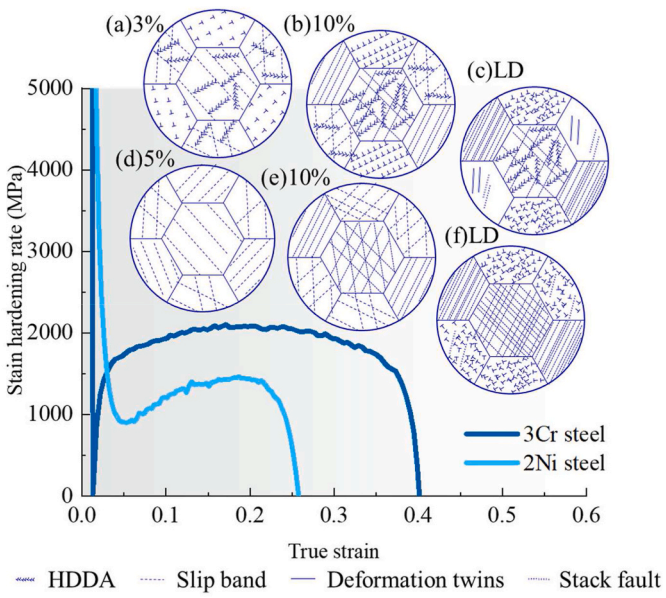


Fig. 12. Schematic diagram showing deformation microstructure at different strain stages in (a–c) 3Cr steel and (d–e) 2Ni steel.

a decrease in the strain hardening rate. When entering strain-hardening stage II (Fig. 6d–f), plastic deformation is characterized by the appearance and proliferation of HDDA (Fig. 6e–f and Fig. 12b). With a further increase in strain, the microstructural evolution is characterized by mutual annihilation and cross-slip through sliding dislocations (Fig. 6e–f and Fig. 12b) and by refinement of the slip bands (Figs. 6d and 12b). This corresponds to strain-hardening stage III (Fig. 4c), where the strain-hardening rate decreases with increasing strain and is dominated by a high density of dislocations (Figs. 6g and 12c) and very fine slip bands (Figs. 6h and 12c). It should also be noted that stacking fault and deformation twins also occur in 3Cr steels (Fig. 7i–l and Fig. 12c), but only at very high strain levels near ultimate failure.

As shown in Fig. 12d–f, the deformation mechanisms of 2Ni steel are mainly mediated by plane slip. In this steel, the slip bands are relatively uniformly distributed (Fig. 7a–b and Fig. 12d), such slip bands multiply with increasing strain, and the space of the slip band decreases (Fig. 7d–f and Fig. 12f). It results that strain hardening rate increase with the strain increasing, and the strain hardening stage is shortened than 3Cr steel. At strain-hardening Stage II, the strain-hardening rate increases with increasing strain. When entering strain-hardening Stage III, it is expected to be due to accelerated dynamic recovery and hard-to-dynamic slip band refinement (Fig. 7i). The dynamic recovery corresponds to dislocations on planar slip bands that are close enough or sufficiently high stresses for dislocation annihilation to occur during cross-slip. A small number of stack faults were found at very high strain levels close to final failure (Fig. 7j–l and Fig. 12e). Overall, Cr alloying of Fe–Mn–Al–C steels promotes the DSBR and HDDA effects of plastic deformation, which can increase elongation.

5. Conclusion

Focusing on austenitic low-density steels with compositions of Fe–20Mn–9Al-1.5C-xNi-yCr (where x = 0, 2 wt% and y = 0, 3, 5 wt%), this study leverages first-principle computational analysis, precipitation kinetics, deformation, and fracture mechanism investigations to successfully design a material that harmonizes exceptional strength and ductility with superior overall performance. Furthermore, it elucidates the intrinsic connection between microstructural features and macroscopic deformation behavior, with the following specific conclusions:

(1) The Fe–20Mn–9Al-1.5C-xNi-yCr steels exhibit superior YS, UTS, and TEL compared to other steels in the Fe–Mn–Al–C series that benefit from effective Ni and Cr alloying. This alloying strategy mitigates the traditional compromise between YS, UTS, and TEL, as demonstrated by 2Ni3Cr steels with YS exceeding 1100 MPa, UTS surpassing 1200 MPa, and TEL greater than 45%.

(2) The strategic addition of Ni and Cr plays a pivotal role in controlling the precipitation dynamics of nano-sized κ-carbides and LRO domains, optimizing their dimensions for enhanced precipitate-induced strengthening. Ni element adding reduces nucleation energy barriers and facilitates the growth and stabilization of κ-carbides, enhancing their efficacy in strengthening. On the contrary, Cr elevates the nucleation energy barrier.

(3) These improvements are attributed to a synergy of grain boundary strengthening, solid solution strengthening, and precipitation strengthening. Additionally, the progressive evolution of the microstructure under strain leads to the repeated formation and multiplication of HDDA and DSBR duel mechanisms.

(4) The enhancement of both strength and strain hardening properties through Ni and Cr alloying is a key contribution of this study. This approach not only elucidates the mechanisms behind the improved properties but also sets a foundational strategy for designing next-generation austenitic lightweight steels with superior mechanical performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge support from the National Basic Research Program, China (No. 2022YFB3705300), National Natural Science Foundation of China (No.52274380, U23A20613), Key R&D projects in Henan Province (No. 231111230400), and Liaoning Revitalization Talents Program, China (XLYC2007030).

References

[1] Chen S, Rana R, Haldar A, Ray RK. Current state of Fe-Mn-Al-C low density steels. Prog Mater Sci 2017;89:345–91. https://doi.org/10.1016/j.pmatsci.2017.05.002.
[2] Zambrano OA. A general perspective of Fe–Mn–Al–C steels. J Mater Sci 2018;53:14003–62. https://doi.org/10.1007/s10853-018-2551-6.
[3] Hu B, Luo H, Yang F, Dong H. Recent progress in medium-Mn steels made with new designing strategies, a review. J Mater Sci Technol 2017;33:1457–64. https://doi.org/10.1016/j.jmst.2017.06.017.
[4] Yurchenko N, Panina E, Kapustin D, Novikov V, Volosevich D, Klimova-Korsmik O, Salishchev G, Zherebtsov S, Stepanov N. A lightweight intermetallic Al-Cr-Ti medium-entropy alloy with good mechanical properties and oxidation resistance. J Alloys Compd 2023;967:171865. https://doi.org/10.1016/j.jallcom.2023.171865.
[5] Jia Y, Ren C, Wu S, Mu Y, Xu L, Jia Y, Yan W, Yi J, Wang G. Multistage strain-hardening behavior of ultrastrong and ductile lightweight refractory complex-concentrated alloys. J Mater Sci Technol 2023;149:73–87. https://doi.org/10.1016/j.jmst.2022.12.011.
[6] Jia Y, Deng X, Wang Q, Li C, Wang Z. Correlation between microstructure, mechanical properties, and slurry erosion behavior of hot-rolled dual-phase steel. Wear 2024:540–1. https://doi.org/10.1016/j.wear.2024.205266. 205266.
[7] Eskandari M, Najafizadeh A, Kermanpur A, Karimi M. Potential application of nanocrystalline 301 austenitic stainless steel in lightweight vehicle structures. Mater Des 2009;30:3869–72. https://doi.org/10.1016/j.matdes.2009.03.043.
[8] Chen L, Li X, Zhao Y, Wang S, Feng Y. Overview of research and development of high-manganese damping steel with integrated structure and function. Acta Metall Sin 2023;59:1015–26.
[9] Kim S-H, Kim H, Kim NJ. Brittle intermetallic compound makes ultrastrong low-density steel with large ductility. Nature 2015;518:77–9. https://doi.org/10.1038/nature14144.
[10] Zhang J, Hu C, Zhang Y, Li J, Song C, Zhai Q. Microstructures, mechanical properties and deformation of near-rapidly solidified low-density Fe-20Mn-9Al-1.2C-xCr steels. Mater Des 2020;186:108307. https://doi.org/10.1016/j.matdes.2019.108307.

[11] An YF, Chen XP, Mei L, Ren P, Wei D, Cao WQ. Precipitation transformation pathway and mechanical behavior of nanoprecipitation strengthened Fe–Mn–Al–C–Ni austenitic low-density steel. J Mater Sci Technol 2024;174:157–67. https://doi.org/10.1016/j.jmst.2023.03.052.
[12] Chen J, Ren J, Liu Z. Deformation microstructures as well as strengthening and toughening mechanisms of low-density high Mn steels for cryogenic applications. J Mater Res Technol 2021;13:947–61. https://doi.org/10.1016/j.jmrt.2021.05.018.
[13] Peng J, Zhang Z, Huang J, Guo P, Zhou W, Wu Y. The effect of grain size on texture evolution and mechanical properties of an AZ31 magnesium alloy during cold-rolling process. J Alloys Compd 2020;817:153302. https://doi.org/10.1016/j.jallcom.2019.153302.
[14] Zhi H, Li J, Li W, Elkot M, Antonov S, Zhang H, Lai M. Simultaneously enhancing strength-ductility synergy and strain hardenability via Si-alloying in medium-Al FeMnAlC lightweight steels. Acta Mater 2023;245:118611. https://doi.org/10.1016/j.actamat.2022.118611.
[15] Yang MX, Yuan FP, Xie QG, Wang YD, Ma E, Wu XL. Strain hardening in Fe–16Mn–10Al–0.86C–5Ni high specific strength steel. Acta Mater 2016;109:213–22. https://doi.org/10.1016/j.actamat.2016.02.044.
[16] Moon J, Park S-J, Jang JH, Lee T-H, Lee C-H, Hong H-U, Han HN, Lee J, Lee BH, Lee C. Investigations of the microstructure evolution and tensile deformation behavior of austenitic Fe-Mn-Al-C lightweight steels and the effect of Mo addition. Acta Mater 2018;147:226–35. https://doi.org/10.1016/j.actamat.2018.01.051.
[17] Hwang JH, Trang TTT, Lee O, Park G, Zargaran A, Kim NJ. Improvement of strength – ductility balance of B2-strengthened lightweight steel. Acta Mater 2020;191:1–12. https://doi.org/10.1016/j.actamat.2020.03.022.
[18] De Cooman BC, Estrin Y, Kim SK. Twinning-induced plasticity (TWIP) steels. Acta Mater 2018;142:283–362. https://doi.org/10.1016/j.actamat.2017.06.046.
[19] Chen S, Rana R, Haldar A, Ray RK. Current state of Fe-Mn-Al-C low density steels. Prog Mater Sci 2017;89:345–91. https://doi.org/10.1016/j.pmatsci.2017.05.002.
[20] Haase C, Zehnder C, Ingendahl T, Bikar A, Tang F, Hallstedt B, Hu W, Bleck W, Molodov DA. On the deformation behavior of κ-carbide-free and κ-carbide-containing high-Mn light-weight steel. Acta Mater 2017;122:332–43. https://doi.org/10.1016/j.actamat.2016.10.006.
[21] Yao MJ, Welsch E, Ponge D, Haghighat SMH, Sandlöbes S, Choi P, Herbig M, Bleskov I, Hickel T, Lipinska-Chwalek M, Shanthraj P, Scheu C, Zaefferer S, Gault B, Raabe D. Strengthening and strain hardening mechanisms in a precipitation-hardened high-Mn lightweight steel. Acta Mater 2017;140:258–73. https://doi.org/10.1016/j.actamat.2017.08.049.
[22] Sutou Y, Kamiya N, Umino R, Ohnuma I, Ishida K. High-strength Fe–20Mn–Al–C-based alloys with low density. ISIJ Int 2010;50:893–9.
[23] Gutierrez-Urrutia I. Low density Fe–Mn–Al–C steels: phase structures, mechanisms and properties. ISIJ Int 2021;61:16–25. https://doi.org/10.2355/isijinternational.ISIJINT-2020-467.
[24] Moon J, Ha H-Y, Kim K-W, Park S-J, Lee T-H, Kim S-D, Jang JH, Jo H-H, Hong H-U, Lee BH, Lee Y-J, Lee C, Suh D-W, Han HN, Raabe D, Lee C-H. A new class of lightweight, stainless steels with ultra-high strength and large ductility. Sci Rep 2020;10:12140. https://doi.org/10.1038/s41598-020-69177-7.
[25] Song W, Zhang W, von Appen J, Dronskowski R, Bleck W. κ-Phase formation in Fe-Mn-Al-C austenitic steels: κ-phase formation in Fe-Mn-Al-C austenitic steels. Steel Res Int 2015;86:1161–9. https://doi.org/10.1002/srin.201400587.
[26] Yao MJ, Welsch E, Ponge D, Haghighat SMH, Sandlöbes S, Choi P, Herbig M, Bleskov I, Hickel T, Lipinska-Chwalek M, Shanthraj P, Scheu C, Zaefferer S, Gault B, Raabe D. Strengthening and strain hardening mechanisms in a precipitation-hardened high-Mn lightweight steel. Acta Mater 2017;140:258–73. https://doi.org/10.1016/j.actamat.2017.08.049.
[27] Han KH, Yoon JC, Choo WK. TEM evidence of modulated structure in Fe-Mn-AI-C austenitic alloys. Scripta Metall 1986;20:33–6.
[28] Dey P, Nazarov R, Dutta B, Yao M, Herbig M, Friák M, Hickel T, Raabe D, Neugebauer J. *Ab initio* explanation of disorder and off-stoichiometry in Fe-Mn-Al-C κ carbides. Phys Rev B 2017;95:104108. https://doi.org/10.1103/PhysRevB.95.104108.
[29] Kimura Y, Handa K, Hayashi K, Mishima Y. Microstructure control and ductility improvement of the two-phase γ-Fe/κ-(Fe, Mn)3AlC alloys in the Fe–Mn–Al–C quaternary system. Intermetallics 2004;12:607–17. https://doi.org/10.1016/j.intermet.2004.03.010.
[30] Sato K, Tagawa K, Inoue Y. Age hardening of an Fe-30Mn-9AI-0.9C alloy by spinodal decomposition. Scripta Metall 1988;22:899–902.
[31] Sutou Y, Kamiya N, Umino R, Ohnuma I, Ishida K. High-strength Fe–20Mn–Al–C-based alloys with low density. ISIJ Int 2010;50:893–9. https://doi.org/10.2355/isijinternational.50.893.
[32] Moon J, Ha H-Y, Park S-J, Lee T-H, Jang JH, Lee C-H, Han HN, Hong H-U. Effect of Mo and Cr additions on the microstructure, mechanical properties and pitting corrosion resistance of austenitic Fe-30Mn-10.5Al-1.1C lightweight steels. J Alloys Compd 2019;775:1136–46. https://doi.org/10.1016/j.jallcom.2018.10.253.
[33] Tuan YH, Wang CS, Tsai CY, Chao CG, Liu TF. Corrosion behaviors of austenitic Fe–30Mn–7Al–xCr–1C alloys in 3.5% NaCl solution. Mater Chem Phys 2009;114:595–8. https://doi.org/10.1016/j.matchemphys.2008.10.009.
[34] Yuan X, Zhao Y, Li X, Chen L. Effect of Cr on mechanical properties and corrosion behaviors of Fe-Mn-C-Al-Cr-N TWIP steels. J Mater Sci Technol 2017;33:1555–60. https://doi.org/10.1016/j.jmst.2017.08.004.
[35] Wang CS, Tsai CY, Chao CG, Liu TF. Effect of chromium content on corrosion behaviors of Fe-9Al-30Mn-(3,5,6.5,8)Cr-1C alloys. Mater Trans 2007;48:2973–7. https://doi.org/10.2320/matertrans.MER2007138.
[36] Frommeyer G, Brüx U. Microstructures and mechanical properties of high-strength Fe-Mn-Al-C light-weight TRIPLEX steels. Steel Res Int 2006;77:627–33. https://doi.org/10.1002/srin.200606440.
[37] Wu ZQ, Ding H, Li HY, Huang ML, Cao FR. Microstructural evolution and strain hardening behavior during plastic deformation of Fe–12Mn–8Al–0.8C steel. Mater Sci Eng, A 2013;584:150–5. https://doi.org/10.1016/j.msea.2013.07.023.
[38] Wang X, Yan M. Effect of cobalt and nickel on the structural stability for Fe3C: first-principles calculations. Int J Mod Phys B 2009;23:1135–40. https://doi.org/10.1142/S0217979209060580.
[39] Moon J, Park S-J, Jang JH, Lee T-H, Lee C-H, Hong H-U, Han HN, Lee J, Lee BH, Lee C. Investigations of the microstructure evolution and tensile deformation behavior of austenitic Fe-Mn-Al-C lightweight steels and the effect of Mo addition. Acta Mater 2018;147:226–35. https://doi.org/10.1016/j.actamat.2018.01.051.
[40] Haase C, Zehnder C, Ingendahl T, Bikar A, Tang F, Hallstedt B, Hu W, Bleck W, Molodov DA. On the deformation behavior of κ-carbide-free and κ-carbide-containing high-Mn light-weight steel. Acta Mater 2017;122:332–43. https://doi.org/10.1016/j.actamat.2016.10.006.
[41] Moon J, Ha H-Y, Park S-J, Lee T-H, Jang JH, Lee C-H, Han HN, Hong H-U. Effect of Mo and Cr additions on the microstructure, mechanical properties and pitting corrosion resistance of austenitic Fe-30Mn-10.5Al-1.1C lightweight steels. J Alloys Compd 2019;775:1136–46. https://doi.org/10.1016/j.jallcom.2018.10.253.
[42] Herrmann J, Inden G, Sauthoff G. Deformation behaviour of iron-rich iron-aluminum alloys at low temperatures. Acta Mater 2003;51:2847–57. https://doi.org/10.1016/S1359-6454(03)00089-2.
[43] Moszner F, Gerstl SSA, Uggowitzer PJ, Löffler JF. Atomic-scale characterization of prior austenite grain boundaries in Fe–Mn-based maraging steel using site-specific atom probe tomography. Acta Mater 2014;73:215–26. https://doi.org/10.1016/j.actamat.2014.04.009.
[44] Liang X, McDermid JR, Bouaziz O, Wang X, Embury JD, Zurob HS. Microstructural evolution and strain hardening of Fe–24Mn and Fe–30Mn alloys during tensile deformation. Acta Mater 2009;57:3978–88. https://doi.org/10.1016/j.actamat.2009.05.003.
[45] Mousavi Anijdan SH, Sabzi M, Najafi H, Jafari M, Eivani AR, Park N, Jafarian HR. The influence of aluminum on microstructure, mechanical properties and wear performance of Fe–14%Mn–1.05%C manganese steel. J Mater Res Technol 2021;15:4768–80. https://doi.org/10.1016/j.jmrt.2021.10.054.
[46] Kim J-K, De Cooman BC. Stacking fault energy and deformation mechanisms in Fe-xMn-0.6C-yAl TWIP steel. Mater Sci Eng, A 2016;676:216–31. https://doi.org/10.1016/j.msea.2016.08.106.
[47] Lan P, Zhang J. Tensile property and microstructure of Fe-22Mn-0.5C TWIP steel. Mater Sci Eng, A 2017;707:373–82. https://doi.org/10.1016/j.msea.2017.09.061.
[48] Springer H, Raabe D. Rapid alloy prototyping: compositional and thermo-mechanical high throughput bulk combinatorial design of structural materials based on the example of 30Mn–1.2C–xAl triplex steels. Acta Mater 2012;60:4950–9. https://doi.org/10.1016/j.actamat.2012.05.017.
[49] Rajasekhara S, Ferreira PJ, Karjalainen LP, Kyröläinen A. Hall–Petch behavior in ultra-fine-grained AISI 301LN stainless steel. Metall Mater Trans A 2007;38:1202–10. https://doi.org/10.1007/s11661-007-9143-4.
[50] Wang SG, Liu SY, Sun M, Tian BH, Zhang ZD. Improved tensile properties, stable microstructures and isotropic deformation of nanocrystalline 304 stainless steel. J Mater Res Technol 2023;23:331–47. https://doi.org/10.1016/j.jmrt.2022.12.197.
[51] Choi JH, Jo MC, Lee H, Zargaran A, Song T, Sohn SS, Kim NJ, Lee S. Cu addition effects on TRIP to TWIP transition and tensile property improvement of ultra-high-strength austenitic high-Mn steels. Acta Mater 2019;166:246–60. https://doi.org/10.1016/j.actamat.2018.12.044.
[52] Miao J, Slone C, Dasari S, Ghazisaeidi M, Banerjee R, George EP, Mills MJ. Ordering effects on deformation substructures and strain hardening behavior of a CrCoNi based medium entropy alloy. Acta Mater 2021;210:116829. https://doi.org/10.1016/j.actamat.2021.116829.
[53] Welsch E, Ponge D, Hafez Haghighat SM, Sandlöbes S, Choi P, Herbig M, Zaefferer S, Raabe D. Strain hardening by dynamic slip band refinement in a high-Mn lightweight steel. Acta Mater 2016;116:188–99. https://doi.org/10.1016/j.actamat.2016.06.037.
[54] Zhang J, Liu Y, Hu C, Jiang Y, Addad A, Ji G, Song C, Zhai Q. The effect of Cr content on intragranular κ-carbide precipitation in Fe-Mn-Al-(Cr)-C low-density steels: a multiscale investigation. Mater Char 2022;186:111801. https://doi.org/10.1016/j.matchar.2022.111801.
[55] Frost HJ, Ashby MF. Deformation-mechanism maps: the plasticity and creep of metals and ceramics. https://api.semanticscholar.org/CorpusID:137438145; 1982.
[56] Honeycombe RWK. The plastic deformation of metals. The plastic deformation of metals 1984;46:145–73. https://doi.org/10.1007/978-3-7091-2848-0_1.
[57] Petch N. The cleavage strength of polycrystals. J. Iron Steel Inst. Lond. 1953;173:25.
[58] Hall EO. The deformation and ageing of mild steel: III discussion of results. Proc Phys Soc B 1951;64:747–53. https://doi.org/10.1088/0370-1301/64/9/303.
[59] Lee J, Kim H, Jeong K, Park S-J, Moon J, Kang S, Han HN. Prediction of precipitation kinetics and strengthening in FeMnAlC lightweight steels. J Mater Res Technol 2021;14:2897–908. https://doi.org/10.1016/j.jmrt.2021.08.094.
[60] Sieurin H, Zander J, Sandström R. Modelling solid solution hardening in stainless steels. Mater Sci Eng, A 2006;415:66–71. https://doi.org/10.1016/j.msea.2005.09.031.
[61] Ardell AJ. Precipitation hardening. Metall Trans A 1985;16:1985–2131. https://doi.org/10.1007/BF02670416.

[62] Meyers M. Mechanical metallurgy: principles and applications. Englewood Cliffs, N. J. 07632, U. S. A: Prentice-Hall, Inc; 1984. p. 761.
[63] Yao MJ, Welsch E, Ponge D, Haghighat SMH, Sandlöbes S, Choi P, Herbig M, Bleskov I, Hickel T, Lipinska-Chwalek M, Shanthraj P, Scheu C, Zaefferer S, Gault B, Raabe D. Strengthening and strain hardening mechanisms in a precipitation-hardened high-Mn lightweight steel. Acta Mater 2017;140:258–73. https://doi.org/10.1016/j.actamat.2017.08.049.
[64] Marquis EA, Seidman DN, Dunand DC. Precipitation strengthening at ambient and elevated temperatures of heat-treatable Al(Sc) alloys. Acta Mater 2003;51:285–7. https://doi.org/10.1016/S1359-6454(02)00496-2.